

Ali Raza

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PROFESSIONAL SUMMARY

Software Engineering student with hands-on experience in machine learning, backend development, REST API development, and real-time web systems. Built end-to-end projects in predictive modeling, web scraping, distributed task processing, and full-stack development using Python, Django, React, PostgreSQL, Redis, and Docker. Seeking an ML Intern, Backend Developer, or AI Intern role to apply strong analytical and software engineering skills.

EDUCATION

Bachelor of Science in Software Engineering

CGPA: 3.68/4.00

The Islamia University of Bahawalpur

6th Semester

Relevant focus: machine learning, backend development, REST APIs, databases, and software engineering fundamentals.

SELECTED PROJECTS

E-Commerce Scraper Engine - Real-Time Price Tracking & Alerts

Django 5 | DRF | React 19 | PostgreSQL | Redis | Celery | Channels | Playwright | Docker Compose

Built a production-grade platform that scrapes dynamic e-commerce pages with BeautifulSoup and Playwright, schedules four-hour price scans with Celery Beat, stores historical prices, and visualizes trends through React and Recharts. Implemented JWT-secured APIs, Axios token refresh, and real-time threshold alerts using Django Channels WebSockets.

ShortIQ - URL Shortener with Real-Time Analytics

Django | DRF | Redis | Celery | PostgreSQL | Docker Compose | JWT

Architected a containerized URL shortening platform with JWT-secured REST APIs, Redis caching, and asynchronous Celery workers. Delivered click analytics with device, browser, and geographic breakdowns plus CSV/PDF export, achieving under 200ms API response times and 10k+ concurrent-user capacity through Docker Compose orchestration.

AI Resume Analyzer - Full-Stack SaaS Platform

Django | DRF | React 19 | PostgreSQL | Google Gemini | scikit-learn | PyMuPDF | Docker

Built an ATS optimization platform combining TF-IDF cosine similarity scoring with Gemini-powered improvement suggestions. Engineered PDF ingestion, matched and missing keyword detection, ReportLab export, and a token-refresh retry queue in React Axios interceptors, deployed as decoupled Docker containers with PostgreSQL persistence.

Bank Customer Churn Prediction

Python | pandas | scikit-learn | XGBoost | SMOTE | Matplotlib

Built an end-to-end ML pipeline to predict bank customer churn. Applied SMOTE for class imbalance, engineered features with ColumnTransformer, and tuned an XGBoost classifier achieving 86.71% test accuracy and ROC-AUC of 0.85 with a 1.41% train-test gap.

Blood Donation Management System

Django | Python | HTML | Oracle | Full-Stack Web Application

Developed a full-stack Django application for managing blood donors, requests, and inventory. Implemented donor registration, blood-group filtering, request tracking, admin management, Django ORM, template rendering, and role-based access control.

Heart Failure Prediction

Python | scikit-learn | pandas | Classification | Feature Engineering | Model Evaluation

Developed binary classification models on structured healthcare data using feature engineering, hyperparameter tuning, and performance benchmarking with precision, recall, F1-score, and ROC-AUC to identify reliable diagnostic signals.

Real-Time Financial Anomaly Detection

Python | scikit-learn | Isolation Forest | PCA | Streamlit | pandas

Built an unsupervised ML pipeline to detect fraudulent credit-card transactions (492 frauds / 284,807 transactions) using Isolation Forest without labeled training. Applied RobustScaler and PCA for two-dimensional anomaly visualization and deployed a real-time Streamlit dashboard for live inference.

EXPERIENCE

Backend Developer Intern
Enigmatix, Bahawalpur

6 months
On-site

Reduced manual processing time by 30% by engineering 5+ Django REST API modules and CRUD systems for business workflows using Python, SQLite, and Oracle.
Improved database query performance by 25% by designing and optimizing Django ORM models, enhancing backend data reliability and response speed.
Cut post-deployment defects by 20% by leading systematic API debugging and enforcing Git/GitHub version-control best practices.
Designed and maintained database models to support scalable backend features and cleaner application logic.

TECHNICAL SKILLS

Programming	Python, SQL
Machine Learning	scikit-learn, XGBoost, regression, classification, SMOTE, model evaluation, feature engineering
Data Analysis	pandas, NumPy, Matplotlib, EDA, data cleaning, feature engineering
Backend	Django, Django REST Framework, REST APIs, Django Channels, JWT, CRUD, SQLite, Oracle, PostgreSQL
Web Scraping	BeautifulSoup, Playwright, dynamic-page extraction, scheduled price tracking
Frontend	React 19, Vite, TailwindCSS, Axios, React Router, Recharts
DevOps / Tools	Docker, Docker Compose, Redis, Celery, Git, GitHub, Gunicorn

CERTIFICATIONS

- Machine Learning with Python (IBM)
 - Data Analysis with Python (IBM)
 - Statistics for Data Science with Python (IBM)
- Exploratory Data Analysis for Machine Learning (IBM)
 - Databases and SQL for Data Science (IBM)
 - AI For Everyone (DeepLearning.AI)

LANGUAGES

Urdu • English